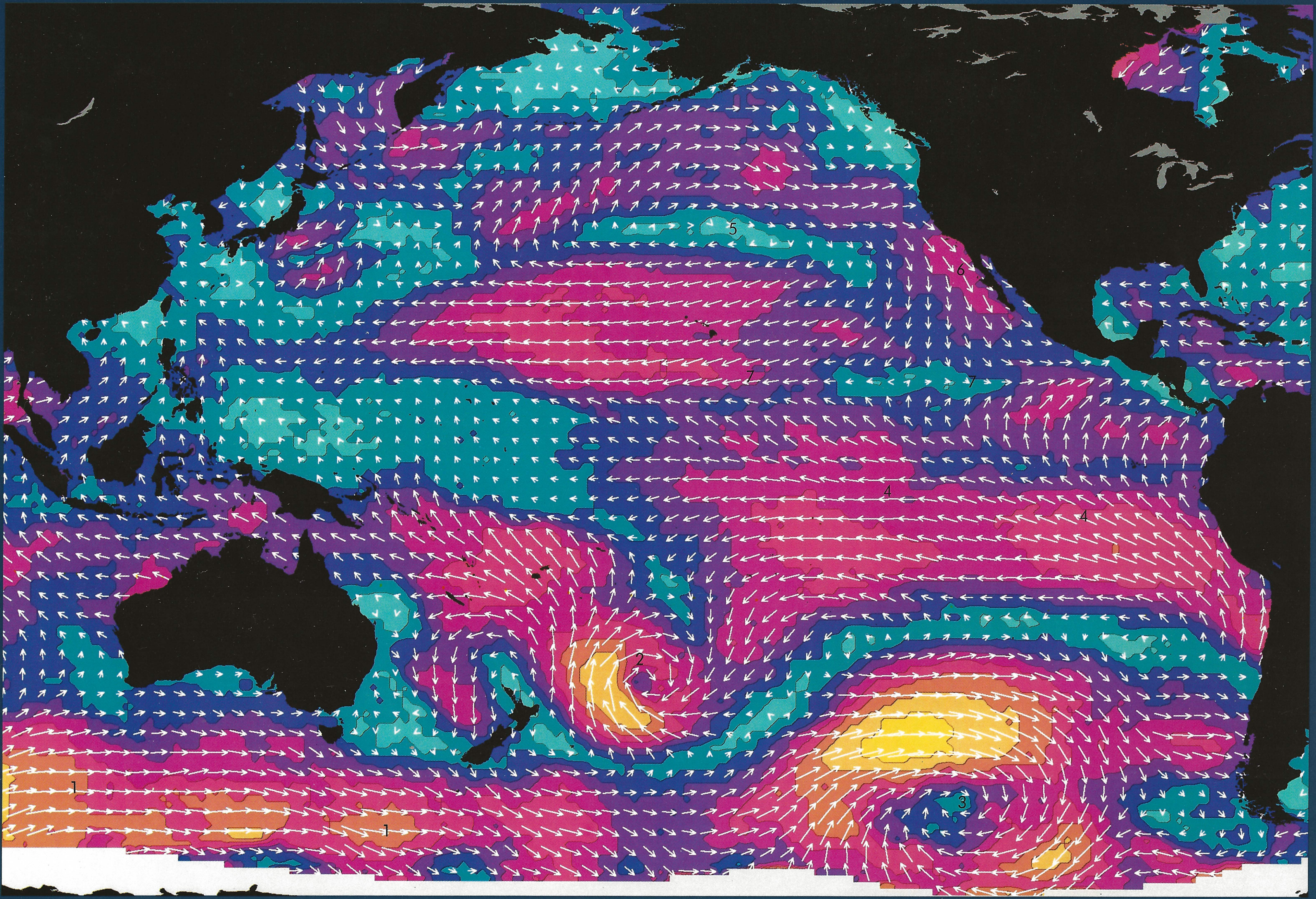
Winds Over the Pacific



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Sea-surface winds, a fundamental driving force for ocean waves and currents, also profoundly influence the exchange of heat between the atmosphere and the ocean. On a global scale the winds and ocean currents are hypothesized to be equal partners in redistributing the excess solar heat gained in the tropics to the cooler polar regions. Thus, knowledge of their dynamics is crucial to understanding our climate.

Our picture of marine winds has come primarily from observations collected by ships. Because ships generally follow well-travelled routes and avoid storms when they can, our knowledge of marine winds has been confined to limited areas. Vast regions of the ocean remain unexplored with no immediate means for routine data collection. This sparseness of wind observations over the ocean has been a significant impediment to studies of both the atmosphere and the ocean.

In 1978, the radar scatterometer (SASS) on board Seasat obtained the first global measurements of winds over the ocean. Pulses of microwave radiation transmitted from the satellite were reflected back to the scatterometer by small wavelets that roughen the sea surface and that arise from the immediate interaction of the sea with the wind. During each orbit, SASS obtained reflections from two 750-km (465-mi) swaths separated by 500 km (310 mi). Complete coverage of the globe required three days. SASS estimated wind speed with an accuracy of ± two meters/second (four miles/hour) and wind direction to within ± 20°.

Displayed here is the Pacific Ocean segment of a global wind field map for September 6 to 8, 1978, produced by Steven Peteherych (Atmospheric Environment Service, Canada), Peter Woiceshyn (Jet Propulsion Laboratory), and Morton Wurtele (University of California, Los Angeles). The arrows point in the direction the winds blow, with longer arrows indicating greater wind speeds. For emphasis, the wind speed is contoured and color coded. Light winds (less than four meters/second or nine miles/hour) are colored blue, and strong winds (greater than 14 m/s or 31 mph) are colored yellow. The continents are black and the Antarctic pack ice is white.

A total of about 350,000 SASS wind measurements were obtained over these three days, more than 100 times the number of wind observations available from ships and buoys during the same period. Although the original SASS wind measurements had a spatial resolution of 50 km (30 mi), the data are displayed here with a coarser resolution. The arrows or wind vectors have a resolution of 3° latitude by 3° longitude, and the wind speed contours correspond to 1° by 1°.

The huge basin of the Pacific Ocean dominates the Earth, covering nearly 40% of its surface area. This SASS wind field for the Pacific presents an opportunity to see atmospheric flow patterns relatively undistorted by the influence of continents. Winter has almost ended in the Southern Hemisphere, and the strong westerlies of the "Roaring Forties" (1), winds blowing from the west, form an almost continuous band over the entire Southern Ocean, interrupted only near the tip of South America. (A glance at the winds over the Atlantic and Indian Oceans will show that these westerlies continue around the globe.) These winds are the strongest, on average, and occur in a region almost devoid of ship traffic.

Two intense cyclonic storms dominate the South Pacific: one off New Zealand (2) and the other closer to South America (3). In the Southern Hemisphere, the flow around a low-pressure or cyclonic storm system is clockwise. In the Tropics, north of 20° S, the southeast Trade Winds (4) extend across much of the South Pacific with speeds of about 12 m/s (27 mph).

Summer in the Northern Hemisphere is characterized by a large high-pressure or anticyclonic system, the Hawaiian High, which has a clockwise rotation about its center near 35° and 150° W (5). The eastern edge of the anticyclone provides strong flow from the north parallel to the coast of southern California and the Baja Peninsula (6). These winds induce the nearshore upwelling important to the coastal fisheries. (See: West Coast Upwelling and Seasonal Baja Ocean Color.)

The northeast Trade Winds constitute the southern flank of the Hawaiian High. At approximately 10° N, the southeast and northeast Trade Winds meet in the Intertropical Convergence Zone (ITCZ) (7), where rising moist tropical air leads to strong convective cloudiness and tropical squalls.

In 1990, NASA plans to fly the next generation scatterometer (NSCAT) on the Navy Remote Ocean Sensing System (NROSS) satellite. During the NROSS mission, scientists will collect global synoptic wind data to study surface waves, wind-driven ocean circulation, and air-sea interaction. In addition, scientists will be working to incorporate the satellite-measured winds into numerical weather-forecasting models; one consequence of this work is that NSCAT winds could have a significant impact on the safety and economic efficiency of maritime operations.

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